

## IK 539 : Sanskrit and Technology: An Overview

Course Code : IK 539

Course Name : Sanskrit and Technology: An Overview

L-T-P-C : 3-0-0-3

Intended for : BTech/MTech/MS/MSc/MA/Ph.D

Prerequisite : IK 540 Bhagavad-Gītā Part II or equivalent understanding of chapters 7-12.

Mutual Exclusion: None

Approval: 42nd Senate

### Course Contents

- **Interactive Sanskrit Teaching Learning Tools:** (12 hours)
  - Introduction to Interactive Sanskrit Learning Tools, Why Interactive Tools for Sanskrit?
    - \* E-learning, Basics of Multimedia, Web-based tools development, HTML, Web page, etc., Tools and Techniques
- **Standard for Indian Languages (Unicode)** (9 hours)
  - Nature of Devanagari/Brahmni scripts, Concept of Aksharas, Conjuncts, and Script grammar.
    - \* Typing in Devanagari Scripts, Typing Tools and Software
- **Text Processing and Preservation Tools** (12 hours)
  - Text Processing, Preservation Techniques, Text Processing and Preservation Tools, and
    - \* Techniques, Survey
- **Optical Character Reader:** Word generation, word analysis, compound formation, sandhi generation and splitting, sentence analysis, Optical Character Reader (OCR), Applications of OCR for Sanskrit and Indian Languages, Tool and Techniques, Survey. (12 hours)

### Textbooks:

1. NA

### Suggested Reading:

1. Teacher's notes, ppt, and handout
2. Bharti A., R. Sangal, V. Chaitanya, "NL, Complexity Theory and Logic" in Foundations of Software Technology and Theoretical Computer Science, Springer, 1990.
3. E-Content suggested by Teacher
4. Tools developed by Computational Linguistics Group, Department of Sanskrit, University of Delhi, Delhi-110007 available at: <http://sanskrit.du.ac.in>
5. Basic concept and issues of multimedia: <http://www.newagepublishers.com/samplechapter/001697.pdf>



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6. Content creation and E-learning in Indian languages: a model: [http://eprints.rclis.org/7189/1/vijayakumarjk\\_01.pdf](http://eprints.rclis.org/7189/1/vijayakumarjk_01.pdf)
  7. HTML Tutorial - W3Schools:
  8. The Unicode Consortium: <http://unicode.org/>